Alerts and the health page

Above: ARCHITECTURE.md §4.7 (outside the box), §6.1 (when something goes wrong) and §7 (the rule "failures are loud").

Talks to:

- the desk's jobs: every run's result, and the jobs that check sources, facts, programs and the box;
- the data sources §3.5: the scheduled health checks;
- the shared record: fact freshness, and whether it can be reached;
- the backups: a night that is not clean, a failed rehearsal;
- qp: the signals its alert cycles find (SIGNAL-ALERTS.md);
- the box: memory, disk, restarts;
- ntfy, which delivers to the phone, and an outside heartbeat check;
- the desk's home, which serves the health page.

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1. What it is

How the desk says something is wrong, and what is right.

- **The signals:** every way the desk can fail, each judged in one place, with its state (green, amber, red), since when, and the fix.
- **The phone:** a red signal is pushed once, through ntfy, and its recovery is announced once.
- **The health page:** every signal on one page, with its fix.
- **The heartbeat:** an outside check that alerts when the desk itself stops reporting, because a desk that is down cannot say so.

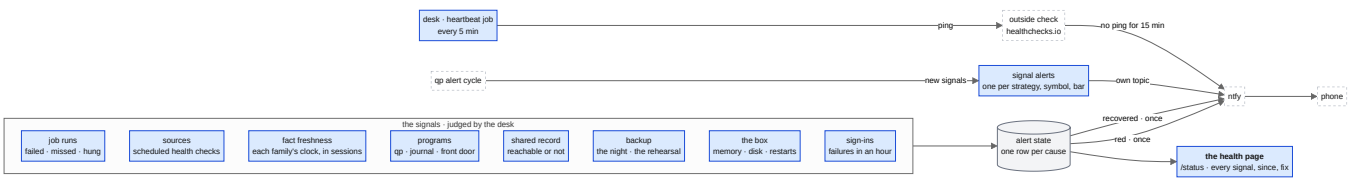
It also carries qp's signal alerts to the phone: a saved strategy firing on one of today's stocks. Those are trading signals, not faults, so they arrive on their own channel.

What it never does:

- repeats an alert while its cause stands;
- stays silent when a job fails or does not run: a silent no-op is a bug;
- sends a secret, a key or a stack trace to the phone;
- lets qp or the journal push on their own: the desk is the only sender.

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2. The diagram



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3. Its parts

3.1 The signals

Each signal has one judge, and amber and red mean the same everywhere:

- **green:** as it should be;
- **amber:** shown on the health page, not pushed. Something to look at;
- **red:** pushed to the phone once, and shown on the health page until it clears.

Signal	Judged by	Amber	Red
a job's run	the scheduler, on every run (JOBS.md)	one failed run of a job that runs more often than every 15 minutes (a scan, a re-quote, the new-ticker run, an alert cycle, a journal sync)	any other job failed, missed or hung; a frequent job failed three runs in a row
a source	the scheduled health checks (DATA-SOURCES.md §3.5), before the day's first scan and at each source's cadence	degraded: some of it failed, or a free tier's limit was reached	down: rejected, empty where it cannot be, or no answer
a fact family's freshness	the desk's freshness job, over `facts_latest`, against each family's clock (CARD.md, RESEARCH-WORKER.md)	older than its clock	older than twice its clock
a program	the desk, asking each program's `/health` every minute	—	qp, the journal or the front door not answering for 3 minutes
the shared record	every program that uses it	unreachable: the tools work from their own files, and the copies wait	unreachable for an hour, or the daily comparison of the copies finds differences
the backup	the night job and the monthly job (BACKUPS.md)	the backup is off (no `data/r2.json`)	a night not clean; the dump did not verify; the rehearsal failed
the box	the desk, every 5 minutes	memory available under 100 MB for 5 minutes; disk under 20 % free	disk under 10 % free; a program restarted by its supervisor (a crash or its memory limit)
sign-ins	the front door (FRONT-DOOR.md)	—	10 failed sign-ins within an hour
the desk itself	the outside heartbeat check (§3.5)	—	no heartbeat for 15 minutes

Counted in trading sessions. A clock that spans a weekend or a holiday is counted on the trading calendar, so Monday morning never reads as stale. The proof: on a Monday before the open, and on the morning after a holiday, no freshness signal is amber.

While a tool moves into the desk (MIGRATION.md step B), the comparison copy also sends one line a day: that day's photo comparison, clean or not.

3.2 The alert state

One row per cause, keyed by what failed (`job:<job>`, `source:<adapter>`, `fact:<family>`, `program:<name>`, `box:disk`, …).

Column	Holds
`key`	the cause
`failing`	1 while it stands
`since`	when it started, or when it recovered
`message`	what failed, and the fix
`sent`	whether the phone was reached (to build)

The rules:

- **Raised once.** The first red raises it and pushes. While it stands, nothing more is sent.
- **Recovered once.** The first good result after it clears it and announces the recovery.
- **Delivered, or tried again.** A push that did not arrive (ntfy unreachable) is tried again at the next judgement while the cause stands (to build: today it is recorded and not tried again).

Where it is kept. A tool's causes are kept in that tool's file; the desk's own causes in the desk's file (FILES.md). The health page reads them all.

3.3 The message

Part	Holds
title	who and what: `T4 · R1 Capture 09:36 failed`, `desk · Polygon down`, `desk · recovered · the night`
body	what failed; since when, in New York time; **the fix**, or where to look (the run's record)
priority	red: high. A recovery or a test: default

Never in a message: a key, a password, an address with credentials, or a stack trace. The stack stays in the log and in the run's record.

3.4 Delivery: ntfy

Setting (`data/keys.json`)	Holds
`alertNtfyTopic`	the topic for faults: a long name only the trader knows
`signalNtfyTopic`	the topic for qp's signal alerts (to build)
`alertNtfyServer`	the ntfy server; `https://ntfy.sh` when not set

The topic is a secret. Anyone who knows it can read the alerts, so it is set with `scripts/set-setting.js` and kept like a key.

Two topics, two channels. Faults and trading signals arrive on different topics, so the trader can silence one without missing the other.

Without a topic, an alert is still written to the log, the alert state and the health page, and the health page says that nothing reaches the phone.

A test: `POST /api/monitor/alerts/test` sends one test alert.

3.5 The heartbeat

The one alert that does not come from the desk, because it is about the desk.

- Every 5 minutes, a desk job pings an outside check (healthchecks.io, free).
- When no ping arrives for 15 minutes, the outside check alerts the phone through its ntfy integration, on the faults topic.
- When the pings resume, it announces the recovery.

What it catches: the desk down or restarting in a loop; the desk's scheduler stuck (the ping is itself a job); the box down; the box's network down.

Its address (`heartbeatUrl` in `data/keys.json`) is a secret like the topic.

3.6 Signal alerts to the phone

When the desk starts one of qp's alert cycles, qp answers with the signals that are new in that cycle (SIGNAL-ALERTS.md).

- The desk pushes each one to the signals topic: the strategy, the symbol, the bar, the price.
- qp's alert history already holds one row per strategy, symbol and bar, so a signal is never pushed twice, even across a restart.
- The page still sounds and notifies, as today.

3.7 The health page

One page, served by the desk's home at `/status`, phone first. It shows every signal of §3.1, grouped:

Group	Shows
the desk	up since; the heartbeat; the last scan of each running tool
jobs	each job of each tool and of the desk: its last run, its result, its next slot, a link to its run history
sources	each adapter's last check and its result, with the fix
facts	each family's age against its clock
programs	qp, the journal, the front door: answering or not, and their version
shared record	reachable or not; how far the copies are behind; the last comparison
backup	the last night, its check, the last rehearsal
box	memory, swap, disk, restarts
delivery	whether alerts reach the phone; a button that sends a test

Each red or amber line says **since when** and **the fix**. The page asks the desk every 30 seconds.

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4. Contracts

It owns	Where
`alert_state`	each tool's file (its causes); the desk's file (the desk's causes)
`health_checks`, each check's result over time	the shared record (DATA-SOURCES.md)

It owns	Where
`alertNtfyTopic`, `signalNtfyTopic`, `alertNtfyServer`, `heartbeatUrl`	`data/keys.json`, mode 0600

It reads	From
every job's runs	`job_runs`, in each tool's file and the desk's file (JOBS.md)
fact ages	`facts_latest`, in the shared record
each program's health	its `GET /health`
new signals	the answer of each alert cycle

Route (served by the desk)	Does
`GET /status`	the health page
`GET /api/status`	every signal, with its state, since when and its fix
`GET /api/monitor/alerts`	the causes failing now, and whether alerts reach the phone
`POST /api/monitor/alerts/test`	one test alert

Only the desk pushes. qp and the journal never send an alert; their faults are seen by the desk's jobs that start them and ask their health.

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5. When it fails

Event	What happens
ntfy is unreachable	the alert is in the log, the alert state and the health page; the push is tried again while the cause stands
no topic is set	nothing is pushed; the health page says so in red
the desk is down	no alert can come from it. The heartbeat check alerts within 15 minutes
a check itself fails (an error in the judge)	that signal turns amber with the error, never green
a cause flaps (fails, recovers, fails)	each change is one push. A job that flaps is itself the fault to fix

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6. Built on

The old desk had no alerts:

- the data check was one button, up to two minutes, judging each source's content only when pressed (legacy/screeners/DATACHECK.md);
- qp's `feedcheck` ran at deploy, and qp's run scripts from a shell;
- a nightly job could fail every night unseen (Q-N5), and the journal's fetch could fail silently (J-4);
- "alpaca ok" was printed daily while the key was being refused.

The design that made it: the old PLATFORM §4, now carried here. qp's signal alert to the phone comes from the old QUANT design ("a push to the phone as well as the page").

The code:

- `src/alerts/push.js`: ntfy, raise once, recover once, the alert state;

- `src/scheduler.js`: failed, missed and hung runs;
- `src/routes/monitor.js`: the alert routes;
- `src/worker/parity.js`: the comparison copy's daily line.

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7. Status

Part	Status
ntfy to the phone; raised once, recovered once; the alert state; the test route	**built**, tested on the phone 2026-10-08
failed, missed and hung runs alert	**built** for R1, R3, the night and the rehearsal. **To build:** every job, with the frequent jobs' three-in-a-row rule (step C)
the comparison copy's daily line	**built** (step 15b)
the scheduled source checks, fact freshness, program health, the box, sign-ins	**to build** (step E)
the health page, `/api/status`	**to build** (step E)
the heartbeat	**to build** (step E)
signal alerts to the phone, on their own topic	**to build** (step C, with the alert cycles started by the desk)
a failed push tried again	**to build** (step E)